

Harvest Models

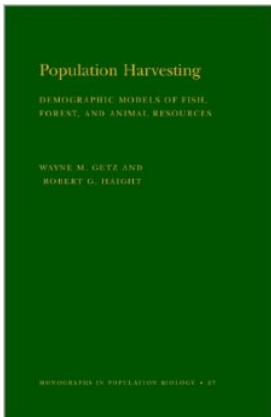


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Sustainable harvest and geometric growth

Sustainable harvest and logistic growth

Definition of maximum sustainable yield (MSY)

Limitations of MSY

Additive vs compensatory mortality

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Sustainable harvest: A harvest that is balanced by population growth such that $N_{t+1} = N_t$

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$$h = r$$

$$N_{t+1} = N_t + N_t r_{max} \left(1 - \frac{N_t}{K} \right) - H_t$$

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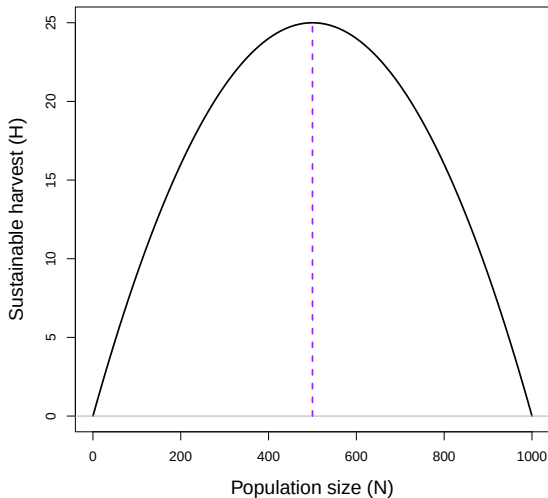
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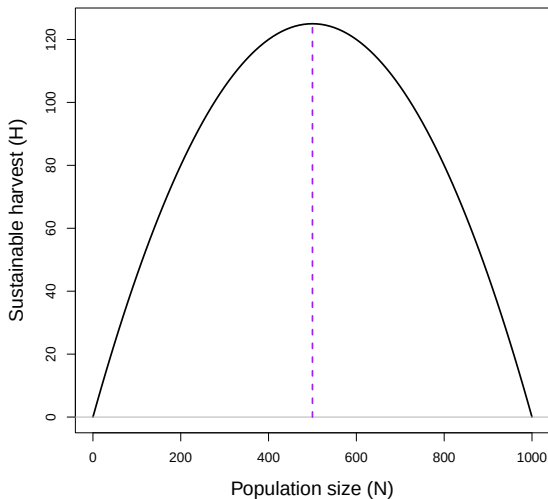
EXAMPLE WHEN $K = 1000$ AND $r_{max} = 0.1$

$$H_t = N_t r_{max} \left(1 - \frac{N_t}{K} \right)$$



EXAMPLE WHEN $K = 1000$ AND $r_{max} = 0.5$

$$H_t = N_t r_{max} \left(1 - \frac{N_t}{K} \right)$$



- MSY is found when $N = K/2$
- The maximum yield is $H = r_{max}K/4$
- The optimal harvest rate is $h = r_{max}/2$

IS MSY USEFUL IN PRACTICE?



PHOTO: STUART GREGORY, GETTY IMAGES

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- Evolutionary consequences?

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- If harvest is compensated for by improved survival, harvest is a form of **compensatory mortality**
- However, if harvest is not compensated for by improved survival, harvest is a form of **additive mortality**

If harvest mortality is additive, extra caution is needed to ensure that harvest doesn't cause long-term population declines.

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COMPENSATORY MORTALITY EXAMPLE

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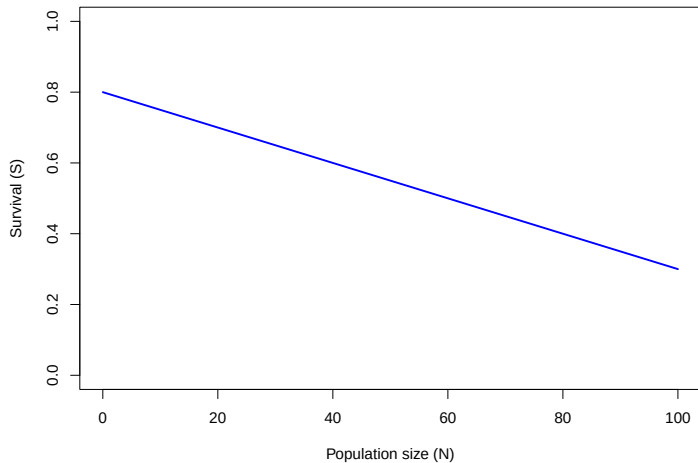
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Natural mortality occurs in a density dependent fashion, such that survival probability (S) declines as N increases.

Let's assume $S = 0.8 - 0.005 \times N$

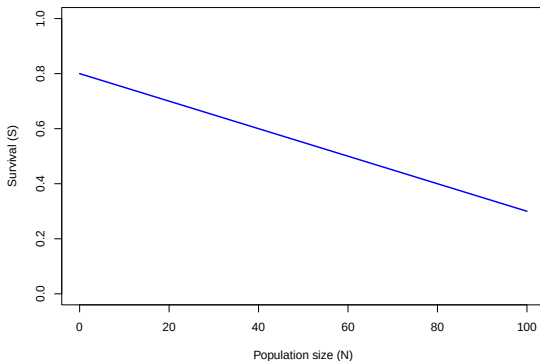
INDIVIDUAL SURVIVAL VS. POPULATION SIZE

$$S = 0.8 - 0.005 \times N$$



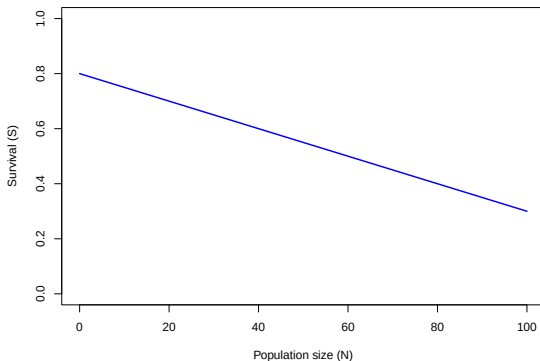
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- Suppose 20 individuals are harvested from an initial population of 100 individuals.



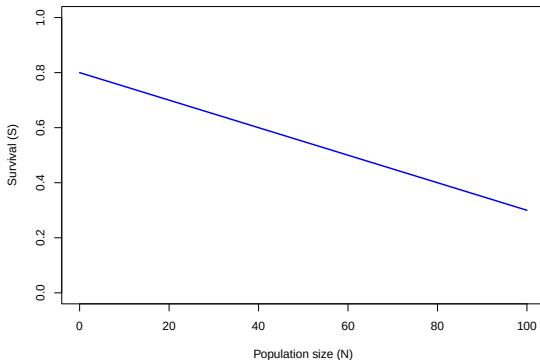
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- Suppose 20 individuals are harvested from an initial population of 100 individuals.
- How many individuals will remain at the end of the year?
- How many would have remained at the end of the year if no hunting had occurred?



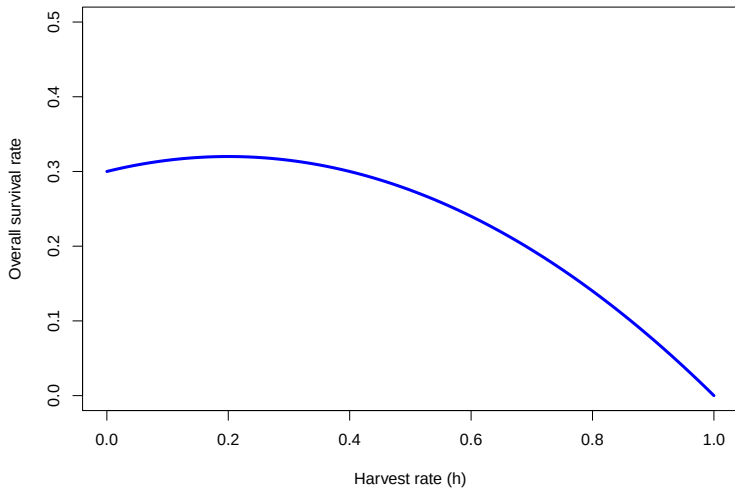
OVERALL SURVIVAL VS. HARVEST RATE

The overall survival rate ($\bar{S}$) is product of survival throughout the hunting season ($1 - h$) and survival after the hunting season

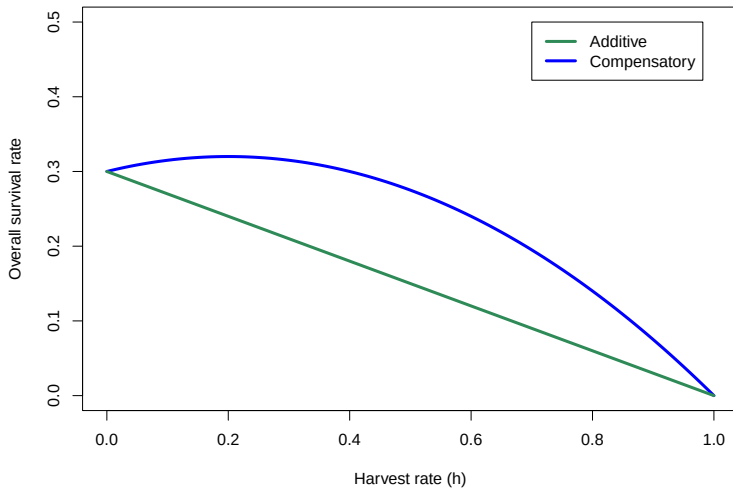
$$\bar{S} = (1 - h)(\beta_0 - \beta_1(N - Nh))$$

OVERALL SURVIVAL VS. HARVEST RATE

$\beta_0 = 0.8$, $\beta_1 = 0.005$, $N=100$

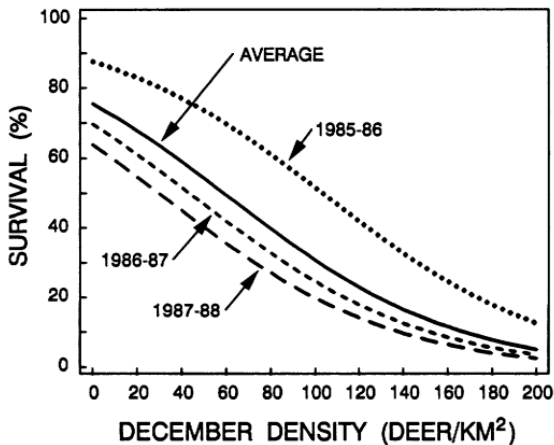


OVERALL SURVIVAL VS. HARVEST RATE



DENSITY-DEPENDENT SURVIVAL EXAMPLE

Mule deer fawn survival (From Bartman et al. 1992)



Key points

- If growth is geometric, sustainable harvest occurs when $h = r$.
- If growth is logistic, maximum sustainable yield occurs at $N = K/2$.
- If survival is density-dependent, harvest mortality can be compensated for by increased survival of remaining individuals (up to a point).
- If mortality is additive, extra caution is needed because harvest is adding to natural mortality without any compensation.
- Managers need to know if harvest mortality is additive or compensatory when setting harvest regulations.

Read pages 22–25 in Conroy and Carroll